

EEE 120

Lab 4 Answer Sheet

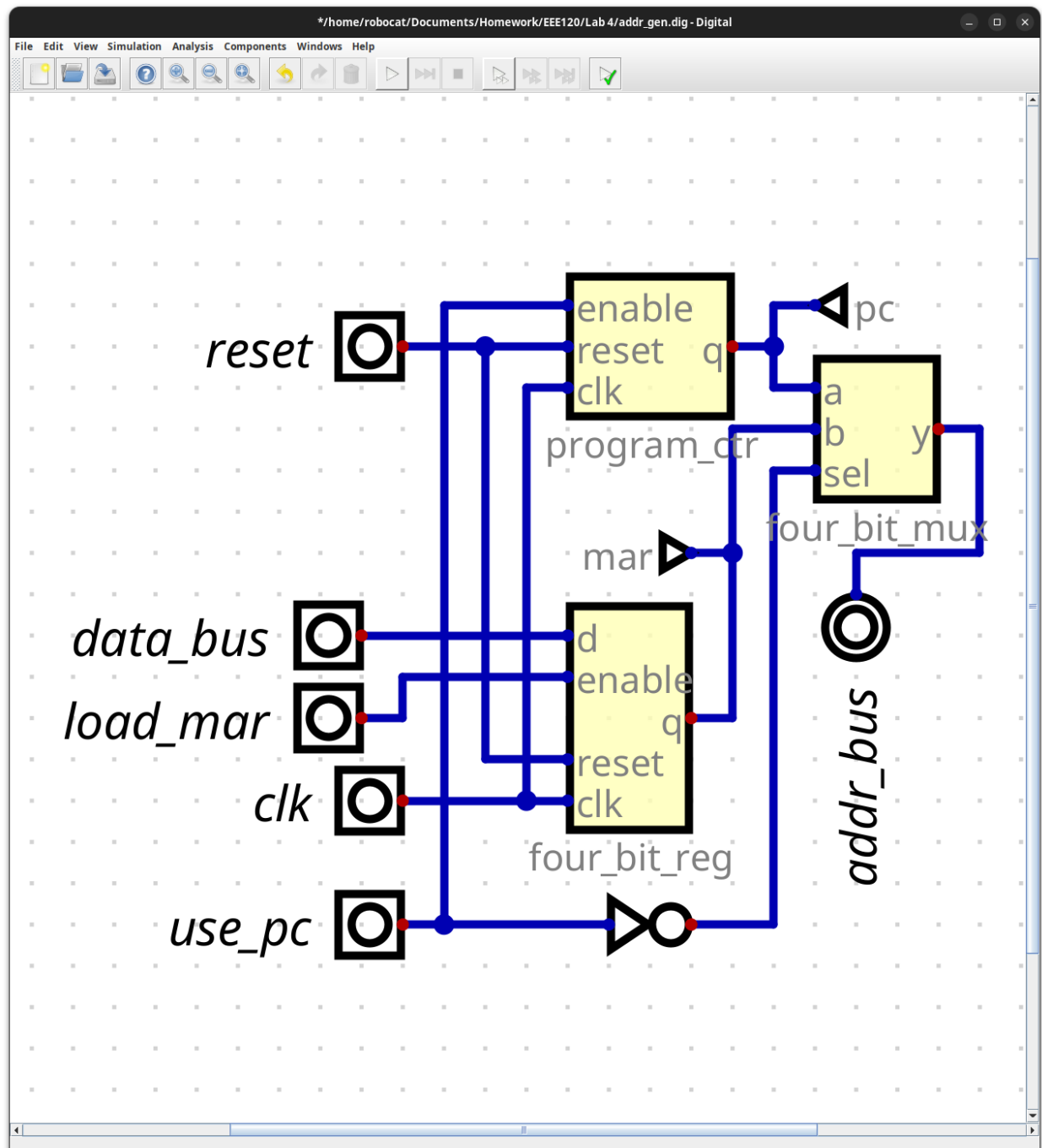
The Complete Microprocessor

Name: Dylan Shah

Instructor/Time: Dr. Steve Millman Tu Th 3:00 PM - 4:15 PM Date: March 26, 2026

Task 4-1: Build and Test the Memory-Address-Generation Circuit

Include a picture of your Digital circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

I did not have any issues designing the circuit.

Did the circuit behave as expected? If no, what was wrong?

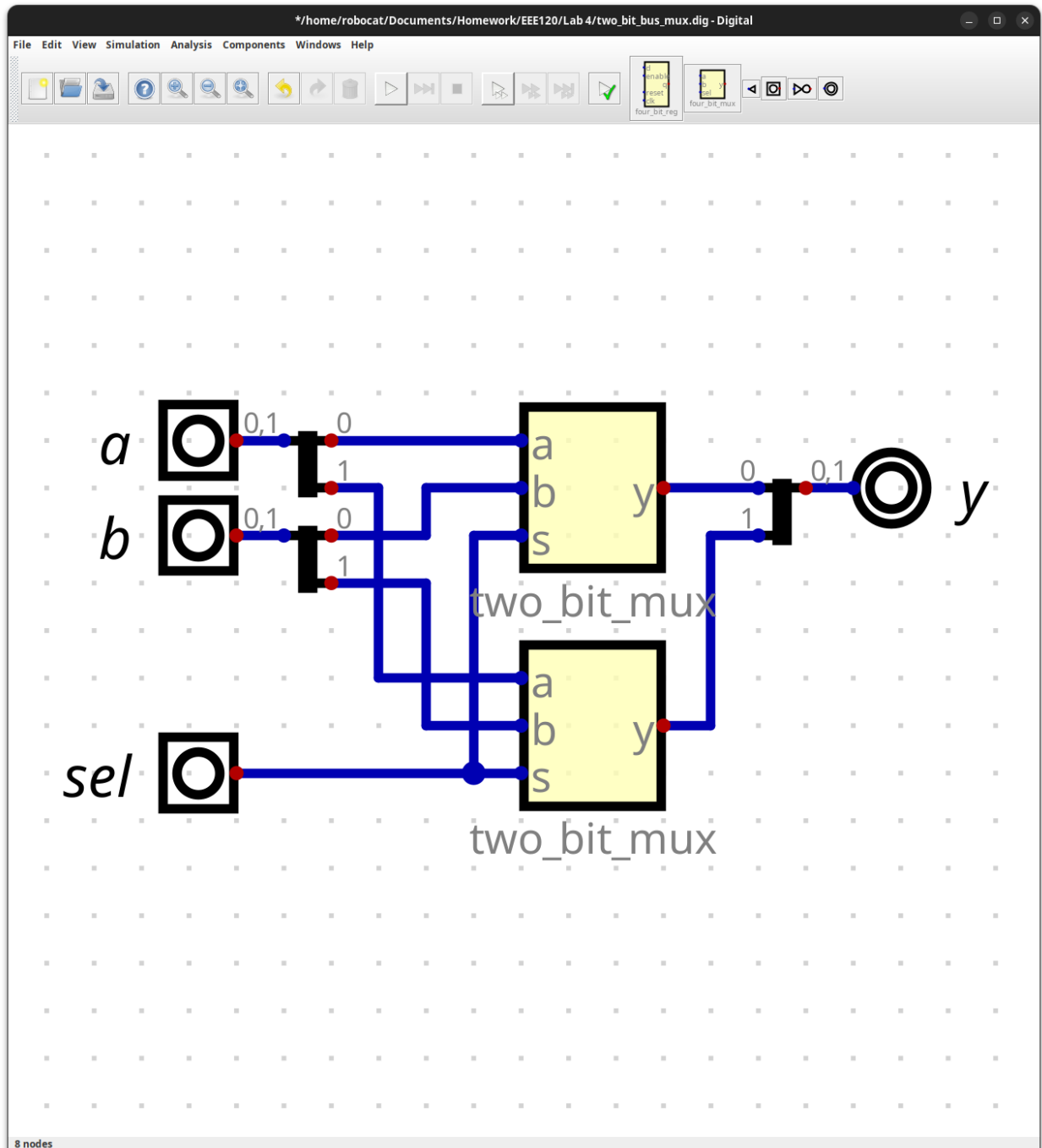
Yes the circuit behaved as I expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I did not realize that the pc should not count when it was disabled, and then I spent a minute trying to look for what was wrong before realizing that it was correct.

Task 4-2: Build and Test the Controller Circuit

Include a picture of your two_bit_bus_mux circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

There were no issues designing the circuit.

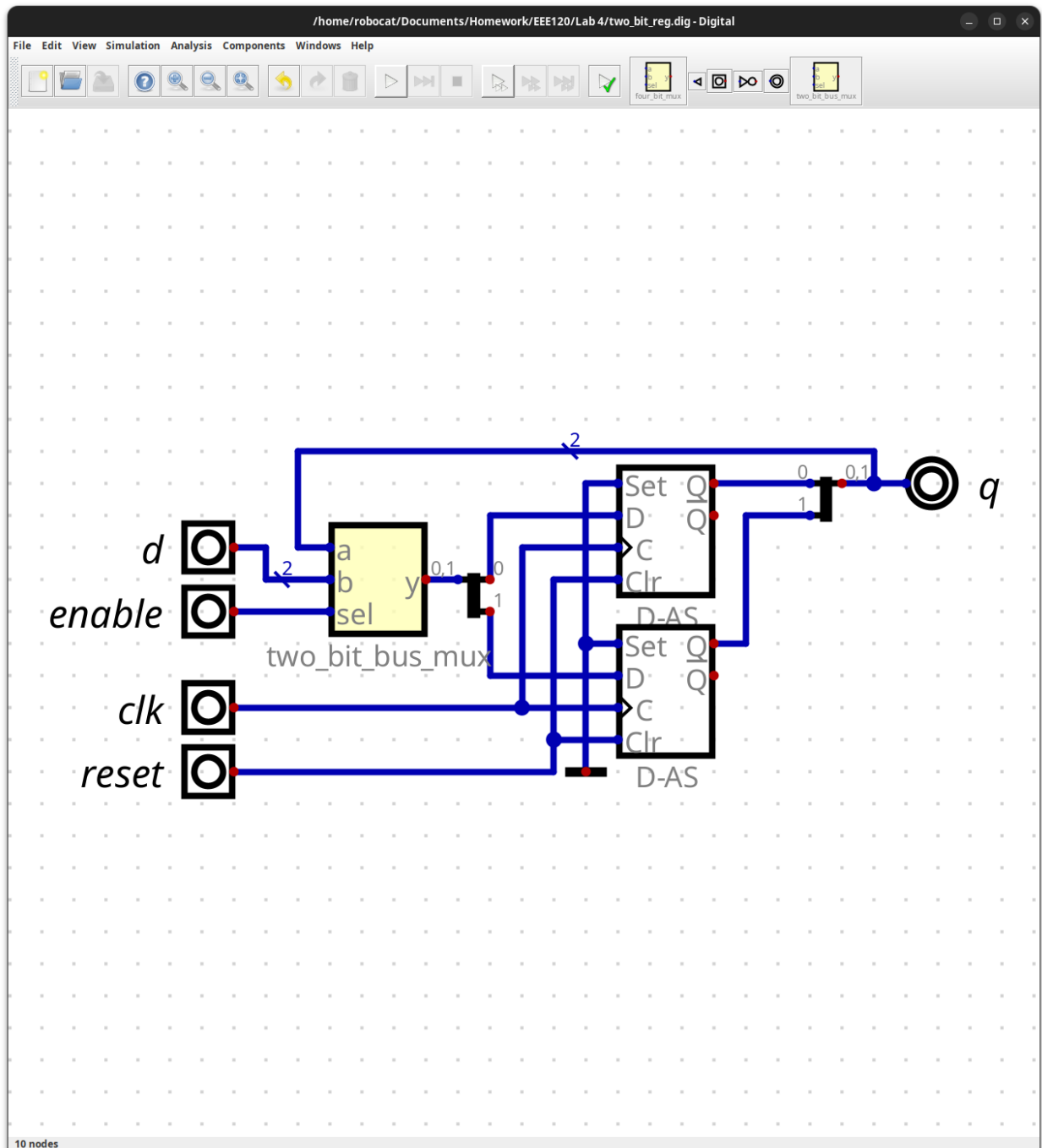
Did the circuit behave as expected? If no, what was wrong?

Yes, the circuit behaved as I expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I did not experience any issues simulating the circuit.

Include a picture of your two_bit_reg circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

I did not have any issues designing the circuit.

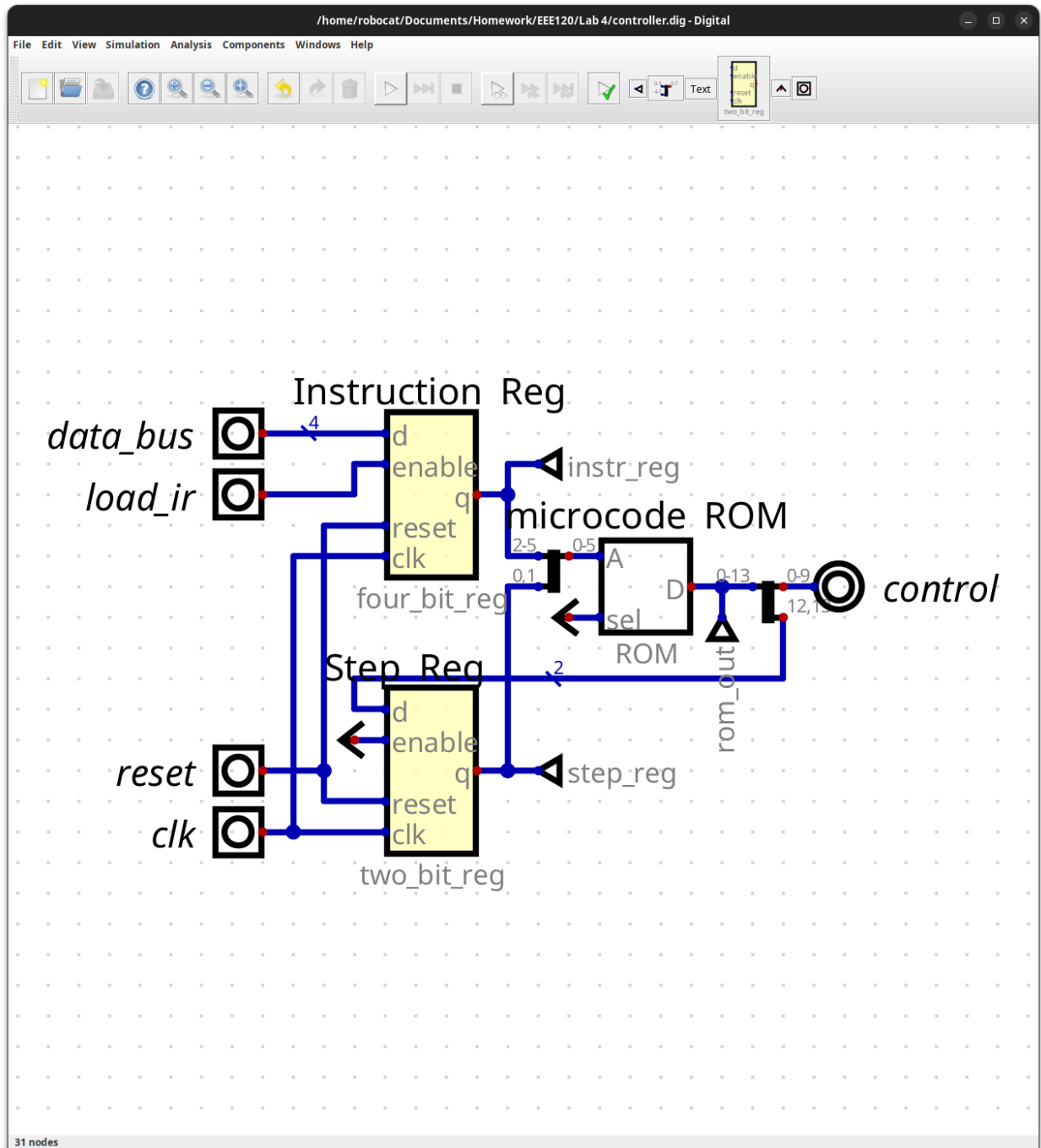
Did the circuit behave as expected? If no, what was wrong?

Yes, the circuit worked as I expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I did not run into any issues simulating the circuit.

Include a picture of your controller circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

It was hard to find the picture of what to build in the instructions.

Did the circuit behave as expected? If no, what was wrong?

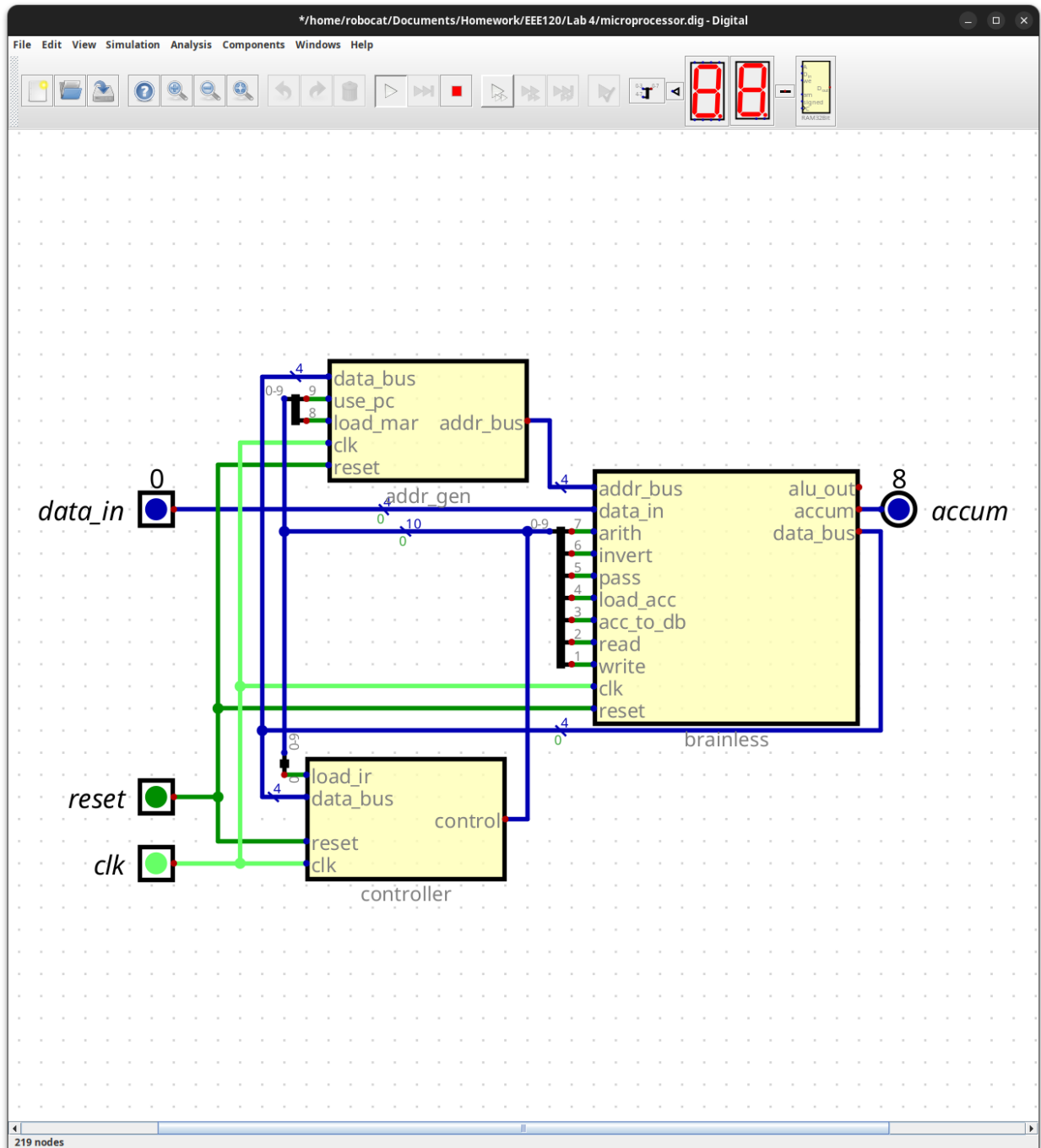
Yes the circuit behaved as I expected it too.

Please comment on the single biggest issue you were facing when simulating the circuit.

It was hard to know what the intended behavior for some input is, so fully testing the circuit is hard.

Task 4-3: Build the Complete Microprocessor Circuit

Include a picture of your Digital circuit here (make sure to show final values as shown in figure 17):



Please comment on the single biggest issue you were facing when designing the circuit.

Getting the splitters to work as I wanted them too.

Did the circuit behave as expected? If no, what was wrong?

Please comment on the single biggest issue you were facing when simulating the processor.

I forgot to replace the # comments with //, leading to my waveform having lots of red x's.+

Did the circuit behave as expected? If no, what was wrong?

Yes, the circuit worked as I expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I did not realize that Digital will not automatically update the ROM like how it updates the RAM, and I had to go in and update it.

Task 4-5: Add the AND, ZERO, SUB, and STORE ACC Instructions

Paste the contents of your final rom_vals here:

v2.0 raw

1205 # LOAD ACC; Load IR

0234 #Load ACC

0000 #unused

0000 #unused

1205 # ADD ACC; Load IR

0294 #ADD

0000 #unused

0000 # unused

1205 # STOP; Load IR

1000 #stay here
0000 #unused
0000 #unused
1205 # AND; Load IR
0214 # AND
0000
0000
1205 # ZERO; Load IR
00D8 # ACC - ACC
0000
0000
1205 # SUB; Load IR
02D4 # SUB
0000
0000
1205 # STORE ACC; Load IR
2304 # Set the MAR to the address
010A # Store var
0000
1205 # Extra Credit; Load IR -- Store Data_in in accum
0030 # Save value
0000
3FFF

4// value to subtract

//After 2 more clocks (80-90ms) accum = -4 = 0xC

6//Store that value in a the param of the next command

B //Address of the `F` below

//After 3 more clocks (110-120ms) mem adress 0xB is now a C

4 //Zero the accum in middle, to show it clears.

//Two clocks later (130-140ms) accum = 0x0

0 // Move the output of the storage command to the acc

F // THIS WILL BE REPLACED BY THE STORAGE COMMAND

//Two clocks later (150-150ms) accum = 0xC

//should be 0xC if set worked, 0xF if failed.

2// STOP

Did the circuit behave as expected? If no, what was wrong?

Yes, the circuit behaves as expected

Please comment on the single biggest issue you were facing when simulating the circuit.

I struggled to think of the order to have the commands to best showcase all the commands.

Task 4-6: Invent Your Own Instruction (Extra Credit)

Place the contents of the rom_vals for the extra credit instruction here:

1205 # Extra Credit; Load IR -- Store Data_in in accum

0030 # Save value

0000

3FFF

Include your Verilog ram_vals.text program used to test the extra credit instruction here:

7 // Store the data on the data bus in accum.

7 // Store the data on the data bus in accum.

7 // Store the data on the data bus in accum.

2 //STOP

test_vals[0] = 12'h0_1_0; // reset - this should always be the first vector

test_vals[1] = 12'h0_0_0;

test_vals[2] = 12'h2_0_2;

test_vals[3] = 12'h2_0_2;

test_vals[4] = 12'hE_0_E;

test_vals[5] = 12'hE_0_E;

test_vals[6] = 12'hA_0_A;

test_vals[7] = 12'hA_0_A;

test_vals[8] = 12'hA_8_A;

Include a picture of your waveforms here:


```

v2.0 raw
1205 # LOAD ACC; Load IR
0234 # Load ACC
0000 # unused
0000 # unused
1205 # ADD ACC; Load IR
0294 # ADD
0000 # unused
0000 # unused
1205 # STOP; Load IR
1000 # stay here
0000 # unused
0000 # unused
1205 # AND; Load IR
0214 # AND
0000
0000
1205 # ZERO; Load IR
0008 # ACC - ACC
0000
0000
1205 # SUB; Load IR
02D4 # SUB
0000
0000
1205 # STORE ACC; Load IR
2304 # Set the MAR to the address
010A # Store var
0000
1205 # Extra Credit; Load IR --
Store Data_in in accum
0030 # Save value
0000
3FFF

```

Task 4-7: Create a video and submit your report

Record a short video showing your schematics in Digital and your waveforms in GTKWave. Be sure to show yourself in the video and show your screen. Explain how your circuit works – you need to convince the grader you did the lab and understand it! **Copy and paste the link to your video below. Make sure the link is working and pointing to the correct video. Remember to include the password if required. Do NOT upload your video to Canvas. It is recommended that you use Zoom to record to the cloud, pasting the link and password below.** If your circuit is not working as expected, explain in the video how it is not working and why you think it is not working.

Video Link: https://drive.google.com/file/d/1Qu1LxkyO8kF20DNQaF-QC-lxZt6VcfVj/view?usp=drive_link

At the beginning of your recording, say your name and the lab name. Be brief in your recording. Submit the completed template to Canvas.

Make sure all your files are in the Lab2 directory. Create a zip file of the Lab2 directory. Remember to turn in the zip file and your completed template on Canvas!

Do not include the video in the zip file! This makes the file very large and you run the risk of the zip file not uploading or taking so long to upload that your submission will be late. Remember that the submission is dated at the time the upload completes, not when it starts!

LAB 4: LAB REPORT GRADE SHEET

Name _____

**NOTE: You submit the zip file in order to show your work.
If the zip file is not submitted you will receive a 0 for this lab!**

Instructor Assessment

Grading Criteria	Max Points	Points Lost
Description of Assigned Tasks. Work Performed &		
Task 4-1: Build and Test the Memory-Address-Generation Circuit	10	
Task 4-2: Build and Test the Controller Circuit	10	
Task 4-3: Build the Complete Microprocessor	10	
Task 4-4: Write and Execute a Simple Program for Your Microprocessor in Simulation	10	
Task 4-5: Add the 'AND', 'Zero', 'Subtract', and 'Store ACC' Instructions	30	
Task 4-6: Invent Your Own Instruction (Extra	10	
Task 4-7: Record your video	10	
Lab Score (80 points total)	Points	
	Late Lab	
	Lab Score	